JR Validated Environment — AS Module

User Guide for Acceptance Sampling

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1. How to Use This Guide

This guide describes the four acceptance sampling scripts available in the JR Validated Environment. Acceptance sampling is used to decide whether to accept or reject a production lot based on the inspection of a sample. The workflow is:

- Step 1 — Design a plan: use jrc_as_attributes or jrc_as_variables to compute sample size and acceptance criteria.
- Step 2 — Visualise the plan: use jrc_as_oc_curve to inspect the OC curve before committing to production sampling.
- Step 3 — Evaluate the lot: use jrc_as_evaluate to apply the plan to real inspection data and obtain a formal ACCEPT or REJECT verdict.

The two plan types are: attributes (pass/fail counts, simpler but less efficient) and variables (continuous measurements, requires fewer samples but the data must follow a normal distribution). See Section 2 for the quick reference chart.

2. Quick Reference

My situation is…	Script	Section
I need to design a sampling plan — my inspection data will be pass/fail (0/1)	jrc_as_attributes	4.1
I need to design a sampling plan — my inspection data will be continuous measurements	jrc_as_variables	4.2
I want to plot the OC curve for a plan I already have (n, c)	jrc_as_oc_curve	4.3
I have collected lot inspection data and need an ACCEPT/REJECT verdict	jrc_as_evaluate	4.4

3. Understanding Acceptance Sampling

3.1 AQL and RQL

Two quality levels define the plan:

- AQL (Acceptable Quality Level) — the fraction nonconforming at which the lot is considered acceptable. The plan must accept lots of this quality with high probability (probability of acceptance $\geq 1 - \alpha$, where α is the producer's risk).

- RQL (Rejectable Quality Level, also called LTPD or LQ) — the fraction nonconforming at which the lot should be rejected. The plan must reject lots of this quality with high probability (≥ 1 − β, where β is the consumer's risk).

3.2 Producer's Risk (α) and Consumer's Risk (β)

The producer's risk α is the probability of rejecting a good lot (one at or better than AQL). The consumer's risk β is the probability of accepting a bad lot (one at or worse than RQL). Both default to industry-standard values: α = 0.05, β = 0.10.

3.3 Attributes vs Variables

An attributes plan inspects units as pass/fail and counts the number of defectives. A variables plan records a continuous measurement for each unit and compares the sample mean and standard deviation against a critical value k. Variables plans typically require significantly fewer samples because each measurement carries more information than a pass/fail result. However, variables plans require that measurements follow a normal distribution and that a specification limit is known.

3.4 OC Curve

The Operating Characteristic (OC) curve plots Pa (probability of accepting the lot) against p (fraction nonconforming in the lot). A good plan has Pa close to 1.0 at AQL and Pa close to 0 at RQL. The steeper and more S-shaped the OC curve, the better the plan discriminates between acceptable and unacceptable lots.

INFO: The scripts use the hypergeometric distribution when the sample fraction n/N exceeds 10% of the lot (large sample relative to lot size). Otherwise the binomial distribution is used. The hypergeometric calculation is more accurate for small finite lots.

4. Per-Script Reference

4.1 jrc_as_attributes — Attributes Sampling Plan (Single and Double)

When do I use this?

Use when your lot inspection will produce pass/fail results (each unit is either conforming or nonconforming). The script finds the minimum sample size n and acceptance number c for a single sampling plan, and n1, c1, c2 for a double sampling plan. Double sampling reduces the expected number of units inspected (ASN) at the cost of a more complex procedure.

What do I need?

Argument	What to provide
lot_size	Total number of units in the lot (positive integer ≥ 2). Used for hypergeometric correction when the sample fraction exceeds 10%.
aql	Acceptable Quality Level as a fraction strictly between 0 and 1 (e.g. 0.01 = 1%). Must be less than rql.
rql	Rejectable Quality Level as a fraction strictly between 0 and 1 (e.g. 0.10 = 10%). Must be greater than aql.
--alpha <val>	Producer's risk (default 0.05). Probability of rejecting a good lot at AQL.
--beta <val>	Consumer's risk (default 0.10). Probability of accepting a bad lot at

	RQL.

Example

```
jrc_as_attributes 500 0.01 0.10
jrc_as_attributes 1000 0.005 0.05 --alpha 0.05 --beta 0.10
```

What to look for in the output

Check the 'Single Sampling Plan' section for n (sample size) and c (acceptance number). The verdict rule is: sample n units; accept if the number of defectives ≤ c. The 'Double Sampling Plan' section (if found) gives n1, c1, c2 with the ASN benefit. The OC curve table shows Pa at key quality levels including AQL and RQL. The PNG saves to ~/Downloads/ with a timestamp prefix.

TIP: For N=500, AQL=0.01, RQL=0.10: Single plan gives n=51, c=2. Double plan gives n1=29, c1=0, c2=2, reducing expected sample size to 36.2 units at AQL (vs 51 for single). Use the double plan when minimising inspection cost is important.

4.2 jrc_as_variables — Variables Sampling Plan (k-method, unknown σ)

When do I use this?

Use when your lot inspection will produce continuous measurements (e.g. dimensions, force, electrical resistance) and you know the specification limit(s). The k-method computes a sample size n and acceptability constant k. The lot is accepted if the Q-statistic $(\bar{x} - LSL)/s \geq k$ (one-sided lower) or $(USL - \bar{x})/s \geq k$ (one-sided upper). Two-sided specifications use --sides 2.

What do I need?

Argument	What to provide
lot_size	Total number of units in the lot (positive integer ≥ 2).
aql	Acceptable Quality Level as a fraction strictly between 0 and 1.
rql	Rejectable Quality Level as a fraction strictly between 0 and 1. Must be > aql.
--alpha <val>	Producer's risk (default 0.05).
--beta <val>	Consumer's risk (default 0.10).
--sides <1|2>	1 = one-sided specification (default). 2 = two-sided specification (both LSL and USL must be checked; tighter plan).

Example

```
jrc_as_variables 500 0.01 0.10
jrc_as_variables 500 0.01 0.10 --sides 2
jrc_as_variables 1000 0.005 0.05 --alpha 0.05 --beta 0.10
```

What to look for in the output

The output shows the sample size n and the acceptability constant k. Collect n measurements from the lot, compute the sample mean $\bar{x}$ and sample standard deviation s, then compute $Q = (\bar{x} - LSL)/s$ (lower-sided) or $Q = (USL - \bar{x})/s$ (upper-sided). Accept the lot if $Q \geq k$. The comparison section shows how many fewer units the variables plan requires compared to the equivalent attributes plan.

TIP: For N=500, AQL=0.01, RQL=0.10 (one-sided): n=21, k=1.7608, achieving α=0.0500, β=0.0956. This is a

58.8% reduction from the equivalent attributes plan (n=51). Variables plans are most beneficial when inspection is costly per unit.

NOTE: The k-method assumes measurements follow a normal distribution within the lot. If the distribution is not normal, the risk levels α and β will not be achieved. Run jrc_normality on pilot data before adopting a variables plan.

4.3 jrc_as_oc_curve — OC Curve for Any Attributes Plan

When do I use this?

Use to plot and tabulate the Operating Characteristic curve for any attributes sampling plan defined by sample size n and acceptance number c. You can supply an optional lot size (for hypergeometric correction), AQL, and RQL (to add reference lines to the plot). This script does not design a plan — it visualises one you already have, whether from jrc_as_attributes or from a standard table.

What do I need?

Argument	What to provide
n	Sample size — positive integer. The number of units to inspect.
c	Acceptance number — non-negative integer. Accept the lot if defectives ≤ c. Must be less than n.
--lot-size N	Optional. Total lot size. If n/N > 0.10, hypergeometric distribution is used instead of binomial. Omit for large lots.
--aql <val>	Optional. AQL as a fraction. Adds a vertical AQL marker to the plot and includes AQL in the Pa table.
--rql <val>	Optional. RQL as a fraction. Adds a vertical RQL marker to the plot and includes RQL in the Pa table.

Example

```
jrc_as_oc_curve 51 2 --lot-size 500 --aql 0.01 --rql 0.10
jrc_as_oc_curve 32 1
jrc_as_oc_curve 80 3 --aql 0.02 --rql 0.08
```

What to look for in the output

The terminal shows a Pa table (p, Pa, 1−Pa columns). Pa at AQL should be close to 1 − α (≥ 0.95). Pa at RQL should be close to β (≤ 0.10). The PNG is saved to ~/Downloads/ and shows the OC curve with AQL and RQL dashed lines if supplied.

4.4 jrc_as_evaluate — Apply a Plan to a Lot (ACCEPT / REJECT)

When do I use this?

Use after sampling a production lot to formally apply the sampling plan and record the lot disposition decision. The script reads a CSV of inspection results, applies the plan criteria, and outputs an ACCEPT or REJECT verdict with a timestamped PNG saved to ~/Downloads/. Supports both attributes (pass/fail) and variables (measurement) modes.

What do I need?

Argument	What to provide
data.csv	Path to the inspection data CSV. Attributes mode: columns id and result (0 = conforming, 1 = defective). Variables mode: columns id and value (numeric measurements).
--type <attributes|variables>	Required. Specify the inspection type: attributes or variables.
--n <value>	Optional. Expected sample size. If provided, the script checks the actual row count against n and warns if they do not match.
--c <value>	Attributes mode: acceptance number. Accept if number of defectives ≤ c.
--k <value>	Variables mode: acceptability constant. Accept if Q-statistic ≥ k.
--lsl <value>	Variables mode: lower specification limit. Used to compute $Q_L = (\bar{x} - LSL)/s$.
--usl <value>	Variables mode: upper specification limit. Used to compute $Q_U = (USL - \bar{x})/s$.

Example

```
jrc_as_evaluate ~/data/lot_results.csv --type attributes --n 51 --c 2
jrc_as_evaluate ~/data/measurements.csv --type variables --n 21 --k 1.7608 --
lsl 9.5
jrc_as_evaluate ~/data/measurements.csv --type variables --k 1.8556 --lsl 9.5 -
-usl 11.5
```

What to look for in the output

The terminal shows the verdict (ACCEPT or REJECT) with the key statistics: defective count vs c (attributes) or Q-statistic vs k (variables). The PNG is a visual lot disposition record saved to ~/Downloads/ with a timestamp prefix. Retain the PNG as objective evidence of the lot disposition decision.

IMPORTANT: jrc_as_evaluate applies the plan you specify. It is your responsibility to confirm that the plan parameters (n, c or n, k) match the validated plan from jrc_as_attributes or jrc_as_variables. Mismatched parameters will produce incorrect verdicts.

5. Preparing Your Data

Plan design scripts (jrc_as_attributes, jrc_as_variables, jrc_as_oc_curve) require no data file — all inputs are command-line arguments. The evaluation script (jrc_as_evaluate) requires a CSV file in the standard jrc two-column format.

Script / mode	Required columns	Notes
jrc_as_evaluate (attributes)	id, result	result: 0 = conforming, 1 = defective
jrc_as_evaluate (variables)	id, value	value: continuous measurement

Column names are case-insensitive. Extra columns are ignored. The 'id' column may contain text labels or numbers. For attributes mode, the 'result' column must contain only 0 or 1. For variables mode, the 'value' column must be numeric. The script will exit with a clear error message if required columns are missing.

TIP: Sample data files are provided in repos/as/sample_data/: as_attributes_example.csv (50 units, 3 defectives) and as_variables_example.csv (20 measurements, mean ≈ 10.5). Use these to test the evaluate

script before running it on production data.

6. Statistical Glossary

Term	Definition
AQL	Acceptable Quality Level. The fraction nonconforming at which the lot is 'good' and should be accepted with high probability (1−α).
RQL / LTPD	Rejectable Quality Level (also Lot Tolerance Percent Defective). The fraction nonconforming at which the lot should be rejected with high probability (1−β).
Producer's risk (α)	Probability of rejecting a lot at AQL quality. The probability of a false rejection. Default: 0.05 (5%).
Consumer's risk (β)	Probability of accepting a lot at RQL quality. The probability of a false acceptance. Default: 0.10 (10%).
n	Sample size. Number of units drawn from the lot for inspection.
c	Acceptance number (attributes plan). Accept the lot if the count of defectives ≤ c; reject if > c.
k	Acceptability constant (variables plan). Accept the lot if the Q-statistic ≥ k.
Q-statistic	For a one-sided lower spec: Q_L = (x − LSL) / s. For a one-sided upper spec: Q_U = (USL − x) / s. Accept if Q ≥ k.
OC curve	Operating Characteristic curve. Plots Pa (probability of accepting the lot) vs p (lot fraction nonconforming). Summarises plan performance at all quality levels.
Pa	Probability of Acceptance. The probability that the sampling plan will accept a lot with fraction nonconforming p.
ASN	Average Sample Number. For double sampling, the expected number of units inspected per lot. ASN < n of the equivalent single plan at AQL.
Double sampling	A two-stage plan: inspect n1 units; accept immediately if defectives ≤ c1; reject if defectives > c2; otherwise inspect n2 more units and accept if total defectives ≤ c2.
k-method	Variables plan method that works with unknown process sigma. The sample standard deviation s is used in place of σ. Uses the non-central t distribution for exact risk control.